

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MARKING SCHEME

NOVEMBER 2016

PHYSICS

9188/5

- 1 (a) (i) Paint spraying, dust extraction (any one) *interlocking, painting, repair - negative impact/effects* B1
- (ii) Spikes *above building, poles* B1
Above building/long poles ~~B1~~
Link from spike to earth B1
A - diagram showing not labeled (1.1) Max 2
labeled (2.1)
- (iii) Conductor of electricity *carbons conductivity to finish. - uses affect paint work* B1
- (b) (i) Force per unit current length *$B = \frac{F}{I}$ term - magnetic field strength* B1
- (ii) Wire X-Y moves to the right/towards power source *$B = \frac{F}{I}$ term - magnetic field strength* B1 *moves*
- Using Fleming LHR *due to electromagnet. force. magnetic force* B1 *(force goes)*
- (iii) $F = BIL$
- $I = \frac{1.5}{1} = 1.5$ B1 *C₁*
- $F = 2 \times 10^{-3} \times 0.1 \times 1.5$ C1
- $= 3 \times 10^{-4} \text{ N}$ *(1 or 2 s.f.)* A1
- (c) (i) 1. Failure due to sustained stress (below yield stress) at high temperature *weakening* B1
2. Failure due to cyclic stress *looking at amplitude below yield stress* B1
- (ii) - Vibrations in grinding mills, *looking at amplitude* ~~B1~~
- motor engines B1
- plane wings *fan blades* Max 1
- fan blades *(repetitive vibrations cause)*
- (iii) Fatigue resistant *stronger / compression* ~~B1~~ *method made*
There are vibrations *carrying heavy loads* M1 *might be given for strong*
- (d) (i) Max possible velocity attained by emitted electrons B1
- (ii) $\phi = hf - ke_{\text{max}}$ *speed photo electrons*
photo electron effect
- $3.03 \times 1.6 \times 10^{-19} = 6.63 \times \frac{10^{-34} \times 3 \times 10^8}{3.5 \times 10^{-7}}$ C1
- $Ke = 8.35 \times 10^{-20} \text{ J}$ *at least 3 s.f.* A1
- 0.522 eV.
- (iii) Some electrons emitted from atomic layers beneath the metal surface *loose energy in collisions with atoms* B1
other electrons B1

- (e) (i) Time taken for half the radioactive Substance/nuclei to decay

$$\frac{X}{X_0} = e^{\frac{-0.693 \times 2}{13}}$$

$$= 0.899$$

$$\% \text{ loss} = (1 - 0.899) \times 100\%$$

$$= 10.1\% \quad \text{at least 2 s.f.}$$

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

note $\lambda \rightarrow$ decay constant of radioactive substance

B1

C1

C1

A1

2 (a) (i) Steady/streamline *(do not cross / no net cross / correct diagrams)* B1

(ii) Same density in the flow B1

(b) (i) liquid incompressible B1

Conservation of mass *$\dot{m}_1 = \dot{m}_2 \Rightarrow B1$
 $\rho_1 A_1 v_1 = \rho_2 A_2 v_2 \Rightarrow \rho_1 = \rho_2 \Rightarrow B1$*

Mass passing per unit time is the same *$A_1 v_1 = A_2 v_2 = \text{constant} \cdot B1$*

(ii) $A_A V_A = A_B V_B = 5 \text{ cm}^3/\text{s}$

$$V_B = \frac{5}{\pi \left(\frac{5}{2}\right)^2} \quad C1$$

$$V_B = 0.255 \text{ cm/s}$$

*0.00255 m/s = 1
accept 1 s.f.*

A1

$$V_A = \frac{5}{\pi \left(\frac{12}{2}\right)^2} = 0.044 \text{ cm s}^{-1}$$

A1

(iii) $P_A + \frac{1}{2} \rho V_A^2 = P_B + \frac{1}{2} \rho V_B^2$

$$100 \text{ kPa} + \frac{1}{2} 800 \times 0.044^2 = P_B + \frac{1}{2} 800 \times 0.255^2$$

C1

$$P_B = 99974.76 \text{ Pa}$$

Accept 1 s.f.f (100 000)

A1

(iv) Low pressure associated with high velocity

B1

$$\text{Accept} = P_A = P_B$$

4/11 - pressure difference may be due to different velocities

3

(a) $Q = \frac{\text{work}}{\text{potential difference}}$ or $Q = IT$

B1

I = current

T = time

1 Coulomb = Quantity of electric charge that passes a given point when a current of 1 A flows per 1 second

$$1C = 1A \times 1s$$

B1

(b) (i) 1. $W = VIT$

$$2.4 \times 10^5 = 240 \times I \times 2.5 \times 60$$

C1

$$I = 6.67 \text{ A}$$

at least 2 s.f.

A1

2. $Q = It$

$$= 6.67 \times 2.5 \times 60$$

~~C1~~

$$= 1000 \text{ C}$$

A1

(ii) $R = \rho l / A$

$$P = VI = 240 \times 6.667 = 1600 \text{ W}$$

$$R = \frac{V}{I}$$

C1

$$1600 \text{ W} = 6.67^2 \times R$$

$$R = 36 \Omega$$

C1

$$36 = \frac{1.1 \times 10^{-6} \times 2.4 \times 4}{\pi b^2}$$

~~C1~~ C₁

$$D = 0.31 \text{ mm}$$

Accept $3.1 \times 10^{-4} \text{ m}$

A1

(iii) - heat lost to the surroundings

B1

- faulty instruments / systematic error and random error

B1

- non uniformity of diameter

B1

- 4 (a) (i) A physical property that varies with temperature B1
- (ii) - shows a linear/monotonic variation B1
 - shows a large change for a small temperature change ^{temp} B1
HW high sensitivity (suitable over temp range being measured)
 * Any plausible suggestion* - reproducible (reliability)
- (b) 1. Very accurate: use of the null method to measure resistance B1
no current flows
2. Requires time to get to thermal equilibrium/high specific heat capacity B1
- Process of measuring resistance etc (*takes long to measure resistance*) B1
- (c) $\frac{\theta}{100} = \frac{4.872 - 5.168}{5.790 - 5.168}$ C1
- $\frac{\theta}{100} = \frac{X_{\theta} - X_0}{X_{100} - X_0}$ C1
- $\theta = -47.59^{\circ}\text{C}$ *accept 5.f* A1
- (d) - Different thermometers only agree at the fixed points/calibration points B1
 - Linear relationship assumed which is not necessarily correct B1
 - One of the thermometers could be faulty ~~B1~~
 - Experimental errors B1
- Max B3

- time between collisions ^{for very large compared to time of collision}
 → mean separation of particles ^{very greater than diameter of particles}
 → volume of particles is negligible as compared to volume of the gas
 → no intermolecular forces
 → perfectly elastic collisions / identical particles (spheres) ^{Newton's mechanics apply}
 → no + particles of very large
 → random motion of particles

5 (a) (i) Any three assumptions (kinetic theory of gases) B3

(ii) Particles would lose kinetic energy/energy conversion, B1

causing - cooling until there is a change ^{phase change.} ~~(which is not possible)~~ B1
 - gas pressure falls B1

(b) (i) Workdone = area under ^{enclosed} trapezium ABCD

$$= \frac{1}{2}(1+3) \times 10$$
 = substitution C2

$$= 1.11 \times 10^5 \text{ J J}$$

accept 2 S.F. A1

(ii) BC - ^{→ isochoric process (B1)} increase of pressure at constant volume B1
 DA - decrease of pressure at constant volume B1

(iii) Increasing the ~~fuel~~ ^{→ accept on BOD} oxygen ratio ^(increase to oxygen/air ratio) B1
 Any plausible suggestion

- use fuel of higher active ratio
- supply the fuel ^{air} mixture to a higher temp before
- ^{spark plug} change to air

$$\frac{1}{2}(11+37)6 = 1.11 \times 10^5$$